MIL-T-82380(TD)
1 September 1969
SUPERSEDING
3313-536
1 July 1968

MILITARY SPECIFICATION

TRAINING AID; MOCKUP, MAGNETRON

This specification has been approved by the Naval Training Device Center, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for an enlarged scale cutaway mockup of a 4J52A Magnetron, mounted on a base suitable for table top use as a training aid.

1.1.1 This item has been designated by the Naval Training Device Center as Device 29HA17 for Navy procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, wood, Cleated-plywood
PPP-B-621	Boxes, wood, Nailed and Lock-corner

Military

MIL-P-116	Preservation; Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-T-17171	Table Top, Plastic; Thermosetting Resin
MIL-T-23991	Training Devices, Military; General Specification for

FSC 6910

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

Numbers 3313-64-410 through 3313-64-417	Magnetron Mockup, Device 29HA17

PHOTOGRAPH

Naval Training Device Center (NAVTRADEVCEN)

Number BXA-13381-1-69	Photograph of a Magnetron Mockup Training Aid

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.

3. REQUIREMENTS

3.1 Materials.- Materials for fabrication of the training aid shall be as specified on drawings NAVTRADEVCEN 3313-64-410 through 3313-64-417. Materials not specified on the drawings shall be as specified in 3.1.1 through 3.1.5.2.

3.1.1 Metals.- Metals shall be in accordance with the metals requirements of MIL-T-23991.

3.1.2 Wood.- Wood shall be select grades of birch or maple having good appearance and finishing qualities, suitable for either clear or opaque finishes. Wood shall be properly seasoned, free from checks, splits, cross grain or seasoning defects which may cause warping.

3.1.3 Plywood.- Plywood shall have a minimum of five (5) plies. Outer surfaces shall be hardwood and all plies shall be free of voids.

3.1.4 Plastics.- Plastics shall be selected to meet the requirements of 3.1.4.1 and 3.1.4.2.

3.1.4.1 Plastics used for components in the sectionalized mockup shall be durable, resistant to cracking, brittleness, discoloration and deformation when subject to the thermal extremes specified in 3.3.

3.1.4.2 Plastic used in the training aid base shall be thermosetting resin conforming to MIL-T-17171, Grade II.

3.1.5 Adhesives.- Adhesives shall be selected to meet the requirements of 3.1.5.1 and 3.1.5.2.

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3.1.5.1 Wood glue.- Wood glue shall be a type not adversely affected by moisture, lacquer or paint.

3.1.5.2 Cements.- Cements for joining plastics or for bonding plastics to other materials shall be those cements recommended by the manufacturers of the plastics for strong homogeneous bonds that will not be adversely affected by the environmental conditions of 3.3. The surface to be cemented shall be thoroughly softened by means of a solvent so that a soft layer or cushion is produced. The bond shall be effected by the uniting and hardening of the softened material. Bonding by capillary attraction shall require specific written approval of the procuring activity for each specific variance.

3.2 Design.- The training aid shall be a visually accurate sectionalized mockup of an operational 4J52A Magnetron at a scale of two to one. The training aid shall be sectionalized as shown on the drawings of 3.1 to enable students to visualize the construction and internal configuration of the operational magnetron. Components shall be color coded to enable students to trace the cathode and heater circuits and the cathode and heater return circuits.

3.3 Reliability.- The training aid shall be capable of withstanding handling by students for a period of two hundred hours per year for three years and shall not be adversely affected in any manner during such handling, storage and transportation under ambient conditions of temperature ranging from minus thirteen to plus one hundred and sixty-five degrees Fahrenheit, and of relative humidity up to ninety-five percent. The device shall be constructed so that no fixed part, subassemblies or assemblies shall become loosened or free under such handling and storage conditions.

3.4 Construction.- The training aid shall be constructed in accordance with the drawings of 3.1. Processes not specified on the drawings shall be in accordance with the process requirements of MIL-T-23991.

3.4.1 Breakage and deformation.- Construction shall be such to minimize the possibility of breakage and deformation during handling by the instructor or students.

3.5 Performance.- The training aid shall be constructed to withstand normal classroom handling and be able to withstand continuous adjusting of the pivotable mount.

3.6 Details of components.- The training aid shall consist of the following components:

(a) Magnetron mockup

(b) Dust cover

3.6.1 Magnetron mockup.- The magnetron mockup shall be a two to one sectionalized scale model of an operational 4J52A magnetron.

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3.6.1.1 The mockup shall be constructed in accordance with 3.4 to show the following components:

(a) Magnet

(b) Wave guide output

(c) Cooling fins

(d) Quarter wave length transformer

(e) Output resonator

(f) Cathode and heater

(g) Cathode and heater return

(h) Choke.

3.6.1.2 A base shall be provided for supporting the mockup during demonstration. The mockup shall be attached to an adjustable bracket to permit inclination and locking in place in various positions selected by students and instructors from the vertical to thirty degrees toward the rear. The adjustable bracket shall be designed to retain the mockup in position if the locking device becomes slightly loosened.

3.6.2 Dust cover.- A dust cover of gray, 16-ounce vinyl-coated nylon, equal to the tensile, tear and flame resistance characteristics of Armor-Tite, shall be provided to protect the training aid when not in use. Covering material used shall be compatible with all finishes of the training aid so that no migrating solvents, contained within the material, will adversely affect any part of the training aid.

3.7 Finish.- All surfaces to be painted shall be properly sealed, primed and then finished with a satin-surface lacquer.

3.7.1 All finished surfaces shall be free from toolmarks, burrs, scratches, sharp edges or any visible imperfections.

3.8 Color.- Colors shall be as indicated on the drawings of 3.1 and shall match the color samples furnished by the procuring agency.

3.9 Nameplates or product markings.- Nameplates and product markings of 3.9.1 and 3.9.2 shall be permanently attached to the training aid for identification.

3.9.1 Nameplate and emblem.- A nameplate and an emblem will be furnished by the procuring agency and shall be affixed to the training aid in the locations shown in photograph number BXA-13381-1-69.

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3.9.2 Nomenclature plate.- A nomenclature plate shall be provided for identification of the components of 3.6.1.1. Markings on nomenclature plate shall be engraved or photochemically applied and shall be in strong contrast with the background material. Color areas on nomenclature plate shall agree in color and shade with corresponding color areas on the mockup and shall be applied in a manner to assure permanent adherance to the background material.

3.10 Workmanship.- The training aid shall be constructed and finished with particular attention given to neatness, correctness of detail, marking and identification of parts and freedom from sharp edges and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical examinations and testing of materials, subassemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection.- Quality conformance inspection shall include the following:

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on a preproduction sample and shall consist of all the examinations specified in 4.5.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government accepted preproduction sample.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

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(a) Materials

(b) Construction

(c) Details of components

(d) Dimensions

(e) Finish and color

(f) Nameplates or product markings

(g) Workmanship

(h) Preparation for delivery.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring agency.

5.1.1 Level A.- For Level A, each device shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring agency.

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2, or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water.

5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment and storage.

5.3.1 Cushioning materials used shall be compatible with all finishes so that migrating solvents in cements, cushioning or components will not affect any part of the training aid.

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5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The magnetron mockup training aid is intended for use in training military personnel in the internal configuration and operation of the operational 4J52A magnetron.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Procuring activity designation (see 1.1.1)

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2)

(d) Gross weight of shipping container (see 5.2.1).

6.3 Definitions.- Except as indicated in 6.3.1, definitions will be furnished upon written request to the procuring activity.

6.3.1 Mockup.- A three-dimensional training aid designed to represent operational equipment. It may be a scaled or cutaway model and may be capable of disassembly or operational simulation.

CUSTODIAN
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N185

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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